

Problem 1. The numbers $1, 2, \dots, 2^{100}$ are written in a board. In every move, MRS chooses 2 numbers a, b and replaces them with $\frac{ab}{a+b}$. After 100 moves. what will be the final number in the board?

Problem 2. The set of integers $1, 2, \dots, 100$ is written on a board. In a move, we can replace a and b with the single value $ab + a + b$. Find all possibilities for the final number.

Problem 3. The set of integers $1, 2, \dots, 100$ is written on a board. In a move, we can replace a and b by the pair of values $0.8a + 0.6b, 0.6a - 0.8b$. Can we reach the set $2, 3, \dots, 100, 101$ by performing only this operation?

Problem 4. There is a graph given on 100 vertices. In each move, GNNT chooses a cycle $\{a_1, a_2, \dots, a_k\}$ and remove all k edges of the cycle, then add a new node and connect it to all vertices of the cycle. Show that GNNT must stop.

Problem 5. Two numbers a, b are written on a blackboard. In every turn we replace, a and b with $a^2 + b^2$ and $2ab$. What will be the sum after 100 moves?

Problem 6. The numbers $1, 2, 3, \dots, 1024$ are written on a blackboard. They are divided into pairs. Then each pair is wiped off the board and non-negative difference of its numbers is written on the board instead. 512 numbers obtained in this way are divided into pairs and so on. One number remains on the blackboard after ten such operations. Determine all its possible values.

Problem 7. Saad moves around on the lattice points according to the following rules: From point (x, y) he may move to any of the points $(y, x), (3x, -2y), (-2x, 3y), (x + 1, y + 4)$ and $(x - 1, y - 4)$. If he starts at $(0, 1)$ can he get to $(0, 0)$?

Problem 8. The sequence $2, 3, \dots, 101$ is written on a board. In each move, G8 chooses two numbers a, b and replaces them with $(a + \frac{b}{2})$ and $(b - \frac{a}{2})$. Can we make the sequence $1, 2, \dots, 100$ by doing this operation?

Problem 9. (PEAK TWIN TRUST) 200 natural numbers are written in a row. For any two adjacent numbers of the row, the right one is either 9 times the left one or $\frac{1}{2}$ times the left one. Can the sum of all these 200 numbers be equal to 24^{2022} ?

Problem 10. The numbers $1, 2, \dots, 1000$ are written on the board. In a turn, one can erase any two numbers a and b and replace them with ab and $a^2 + b^2$. Prove that it is impossible to get at least 700 identical numbers by performing such operations.

Problem 11. In a line there are 100 poddo – pata (peak pata). The i -th pata has a_i frogs (peak frog). In each minute sad Saad chooses a pata with at least 2 frogs and moves two frogs from the pata to the adjacent patas (both go to different patas). If the chosen pata is at the edge, then one of the frog jumps into the water and gets unalived. Will the game eventually stop (Prove or disprove)?

Problem 12. There is a string of a, b, e . We can replace any two elements with one element in the following way:

$$a, b \rightarrow e$$

$$a, e \rightarrow a$$

$$b, e \rightarrow b$$

$$a, a \rightarrow b$$

$$b, b \rightarrow a$$

$$e, e \rightarrow e$$

Prove that, no matter how we do this, the final letter will be uniquely determined by the initial string.